Application Notes

Bombas Táchira CA | Quotations PF-102126-DB, PF-102126-DB2, PF-102126-DB3 | Sensidyne, LP

System Overview

Quote	Scope
PF-102126-DB	Two SensAlarm FLEX units (general purpose). Unit 1: CO 0-500 ppm at breathing height, H2S 0-10 ppm remote-mounted low. Unit 2: infrared LEL.
PF-102126-DB2	Calibration kit, calibration gases and consumables for the FLEX units.
PF-102126-DB3 (optional)	SensFlex PID transmitter, 0-5 ppm, with horn, strobe, power supplies and PID calibration kit.

Each FLEX unit alarms locally with its built-in horn, strobe and display.

H2S Sensor

- Lowest range available, 0-10 ppm (823-0206-23). Minimum indicated concentration: 0.3 ppm.
- Alarm setpoints must fall within the sensor range. We suggest 5 and 10 ppm instead of 10 and 15 ppm.

Infrared LEL Sensor

- Recommended calibration gas for gasoline and crude oil vapors: pentane at 50% LEL (009824-72).
- This sensor warns of flammable concentrations. It will not respond to odor levels from the port; the PID is the more sensitive option for odors.

PID Sensor (Optional, PF-102126-DB3)

- Range 0-5 ppm (lowest available), 10.6 eV, configured for gasoline. Detects both gasoline and diesel vapors.
- Accuracy: ±1% of full scale. Readings are in ppm; a ppb range is not available.
- Factory configuration: gasoline (CAS 8006-61-9), response factor 0.9, calibrated with 2.5 ppm isobutylene (009824-99), 2.25 ppm span value.

Important: PID sensors are not gas specific. They respond to hundreds of volatile organic compounds. Fuel vapors often contain benzene, toluene, ethylbenzene and xylenes (BTEX). With the sensor configured for gasoline, these compounds read higher than their actual concentration:

Compound	Response factor (10.6 eV)	Approximate reading vs. actual
Gasoline (target)	0.9	1.0x
Benzene	0.5	1.8x
Toluene	0.56	1.6x
Ethylbenzene	0.56	1.6x
Xylenes	0.54	1.7x

The detector may therefore alarm when gasoline is not the cause.

Local Alarm Power (PF-102126-DB3)

- SensFlex transmitter: dedicated 10 W, 24 VDC power supply (821-9904-01).

- Horn and strobe: share a second 10 W, 24 VDC supply. Calculated load approximately 6.1 W (horn 0.8 W, strobe approximately 5.3 W).

Calibration and Maintenance

- FLEX sensors (CO, H2S, infrared LEL): calibrate every 6 months, with bump tests in between. The FLEX can be set to remind you when calibration is due.
- Infrared LEL: zero and span calibration are performed as separate steps.
- PID: we recommend leaving maintenance to Sensidyne through the PID Sensor Exchange Program, which includes four calibrations per year (one every 90 days) and an annual sensor rebuild with parts discounted 25%.
- Replacement PID lamp, for reference: 821-1205-01.

Sensor Life and Replacement Part Numbers

Sensor	Part number	Expected life
H2S 0-10 ppm	823-0206-23	3 years
CO 0-500 ppm	823-0219-22	3 years
Infrared LEL 0-100%	823-0211-51	5 years
PID 0-5 ppm	823-1251-14	4 years / 1 year for lamp

Spare H2S and CO sensors are not included. Electrochemical sensor life is counted from the shipping date and continues while the sensor is in storage, as the electrolyte slowly dries out and the electrode gradually changes. We recommend ordering replacements as each sensor approaches the end of its life, so they arrive fresh.

Warranty

- Instruments: 2 years from shipment
- Sensors: 14 months from shipment
- PID sensor: 12 months from shipment